CEI Company Asked Questions

Interview questions with short, ready-to-use answers

1. How do you attract web users?

Attracting users to a website combines good content, visibility and a smooth experience:

- **SEO:** relevant keywords, clean URLs, meta tags, fast loading and mobile-friendly design so the site ranks higher on search engines.
- **Quality content:** useful blogs, videos and tutorials that solve the visitor's problem.
- **Social media and email marketing:** share content and bring back returning visitors.
- **Good UI/UX:** simple navigation, responsive layout and quick page speed (this is where lazy loading helps).
- **Paid ads and analytics:** use Google Ads for reach and Google Analytics to study user behaviour and improve.

2. Join query on 2 or 3 tables

A JOIN combines rows from multiple tables using a related column (usually a foreign key).

Two tables (Students and Departments):

```
SELECT s.name, d.dept_name
FROM students s
INNER JOIN departments d ON s.dept_id = d.dept_id;
```

Three tables (Students, Departments, Courses):

```
SELECT s.name, d.dept_name, c.course_name
FROM students s
INNER JOIN departments d ON s.dept_id = d.dept_id
INNER JOIN courses c ON s.course_id = c.course_id;
```

Types of joins: INNER (only matching rows), LEFT (all rows from left table), RIGHT (all rows from right table), FULL (all rows from both).

3. Agile: basic technical terms and interview questions

Agile is an iterative way of building software in small increments, with continuous feedback and the ability to adapt to change.

Term	Meaning
Sprint	A fixed time box (usually 1 to 4 weeks) in which a set of work is completed.
Product Backlog	Prioritised list of all features and requirements for the product.
Sprint Backlog	Items selected from the product backlog for the current sprint.
User Story	A short requirement written from the user's view: 'As a user, I want X so that Y'.
Scrum	The most popular Agile framework, with defined roles, events and artifacts.

Term	Meaning
Scrum Master	Facilitates the process and removes blockers for the team.
Product Owner	Represents the customer and decides the backlog priority.
Daily Stand-up	A 15-minute daily meeting: what I did, what I will do, any blockers.
Sprint Review / Retrospective	Review demonstrates the work; retrospective discusses how to improve the process.
Kanban	A visual board (To Do, In Progress, Done) to manage workflow.
Velocity	The amount of work a team completes per sprint.

Common questions:

- **Agile vs Waterfall?** Waterfall is sequential and rigid; Agile is iterative and flexible, with customer feedback after every sprint.
- **What is the Agile Manifesto?** It values individuals and interactions, working software, customer collaboration and responding to change.
- **What is Scrum vs Kanban?** Scrum uses fixed-length sprints and roles; Kanban is continuous flow with work-in-progress limits.
- **Which Agile role would you prefer?** Developer in the team, contributing to sprint goals and daily stand-ups.

4. Application of a data structure is given; identify the data structure

Match the application to the structure:

Application	Data Structure
Undo/Redo, browser back button, function calls, expression evaluation	Stack (LIFO)
Printer queue, CPU scheduling, customer service line, BFS	Queue (FIFO)
Dynamic memory, music playlist, image viewer next/previous	Linked List
Priority scheduling, finding top-k elements, Dijkstra's algorithm	Heap / Priority Queue
File system, HTML DOM, organisation hierarchy	Tree
Dictionary, caching, fast lookup by key	Hash Table
Social networks, maps and GPS routes, network routing	Graph
Autocomplete, spell check	Trie

5. If 10 pages are used, how would you render only the selected page?

Answer: Lazy loading.

Instead of loading all 10 pages at the start, load a page only when the user navigates to it. This reduces the initial load time and memory usage. In React, for example:

```
const Page3 = React.lazy(() => import('./Page3'));

<Suspense fallback={<p>Loading...</p>}>
  <Page3 />
</Suspense>
```

Combined with routing (React Router), only the component for the selected route is rendered and its code is fetched on demand (code splitting).

6. A URL is given; name its parts

Example: https://www.example.com:8080/products/list?category=books&page=2#reviews

Part	Value in example	Meaning
Protocol / Scheme	https	Rule used to communicate
Subdomain	www	Sub-section of the domain
Domain name	example.com	Name of the website
Port	8080	Network port (default 80 for http, 443 for https)
Path	/products/list	Location of the resource on the server
Query string	?category=books&page=2	Key-value parameters sent to the server
Fragment	#reviews	Jumps to a section within the page

7. GET and POST methods

Feature	GET	POST
Purpose	Retrieve data	Send or create data
Data location	In the URL (query string)	In the request body
Security	Less secure; data visible in URL and history	More secure; data not shown in URL
Data size	Limited by URL length	No practical limit
Caching / bookmark	Can be cached and bookmarked	Not cached or bookmarked
Idempotent	Yes	No
Example	Search a product	Login form, signup form

8. Why doesn't Java support multiple inheritance?

Java does not allow a class to extend more than one class to avoid the diamond problem and ambiguity.

If class C extends both A and B, and both A and B define the same method show(), the compiler cannot decide which version C should inherit. This ambiguity makes code complex and error-prone, so Java avoids it for simplicity.

Alternative: a class can implement multiple interfaces, which gives the benefit of multiple inheritance of type without the ambiguity (if two interfaces share a default method, the class must override it explicitly).

9. Process vs Thread

Feature	Process	Thread
Definition	A program in execution	Smallest unit of execution inside a process
Memory	Has its own separate memory space	Shares memory of its parent process
Creation cost	Heavy, slower	Light, faster
Communication	Needs IPC (pipes, sockets)	Easy, via shared memory
Context switch	Slower	Faster
Failure impact	Independent; one crash does not affect others	A thread crash can bring down the whole process
Example	Chrome and VS Code running separately	Tabs or background tasks inside one app

10. Priority scheduling: working and real-life example

In priority scheduling, each process is assigned a priority and the CPU is given to the process with the highest priority first. Processes with equal priority are handled in FCFS order. It can be preemptive (a higher-priority arrival interrupts the running process) or non-preemptive.

Problem: starvation of low-priority processes. **Solution:** aging, which gradually increases the priority of waiting processes.

Real-life example: a hospital emergency room. A critical accident patient is treated before someone with a mild fever, even if the fever patient arrived earlier. Similarly, an ambulance gets priority at a traffic signal.

11. Application of binary search

Binary search finds an element in a sorted collection in O(log n) time by repeatedly halving the search range.

- Searching a name in a sorted contact list or dictionary.
- Finding a word or record in a sorted database index.
- Debugging with git bisect to find the commit that introduced a bug.
- Finding the square root or solving 'guess the number' problems.
- Finding the first or last occurrence, or lower and upper bounds, in a sorted array.

12. Abstraction: code implementation

Abstraction hides implementation details and shows only essential features. In Java it is achieved with abstract classes and interfaces.

```
abstract class Shape {
    abstract double area();          // no implementation
    void display() {                 // concrete method
        System.out.println("Area: " + area());
    }
}

class Circle extends Shape {
    double r;
    Circle(double r) { this.r = r; }
    double area() { return Math.PI * r * r; }
}

class Rectangle extends Shape {
    double l, b;
    Rectangle(double l, double b) { this.l = l; this.b = b; }
    double area() { return l * b; }
}

public class Main {
    public static void main(String[] args) {
        Shape s1 = new Circle(5);
        Shape s2 = new Rectangle(4, 6);
        s1.display();
        s2.display();
    }
}
```

Real-life analogy: when you drive a car you use the steering wheel and pedals without knowing how the engine works internally.

13. Where do you see yourself in 5 years?

In five years I see myself as a skilled and dependable professional in this field, having built strong technical depth and taken on greater responsibility. In the first couple of years I want to learn the company's technologies and processes and contribute meaningfully to my team. After that I hope to lead small modules or projects and mentor newer members, while growing alongside the company.

14. What is your ambition?

My ambition is to become an expert in AI and software development who builds solutions that solve real problems. I want to keep learning new technologies, work on impactful projects and eventually grow into a role where I can lead a team and add value to the organisation.

15. Favourite subject and least favourite subject

Favourite subject: Transfer Learning (TL).

I enjoy Transfer Learning because it reuses a model trained on a large dataset (such as ImageNet) and adapts it to a new task with much less data, time and computing power. It showed me how practical AI can be built efficiently, and it made me curious to explore pre-trained models and fine-tuning further.

Least favourite subject: Hindi / R Programming.

I found R Programming less engaging because its fundamentals (variables, loops, functions, data structures) are the same as Python, which I had already learned, so I did not gain much that was new. I still understood its strengths in statistics and data visualisation, and it helped me see how the same concepts transfer across languages.

Tip: keep the 'least favourite' answer positive. Mention what you learned or why it was less challenging, not that you disliked it.